Recall. X compact Kähler (e.g. smooth proj var)

$$\Rightarrow H^k(X, \mathbb{C}) \cong \bigoplus_{p+q=k} H^q(X, \Omega_X^p)$$

This puts a "pure Hodge structure" on the sing. coh of X:

Def^n. A <u>rational pure Hodge structure</u> of weight k consists of

- A finite-dim vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$
- A decomposition

$$H_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C} = \bigoplus_{p+q=k} H^{p,q}$$

$H_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$ denote by $H_{\mathbb{C}}$

where $\overline{H^{p,q}} = H^{q,p}$

(Since $H^{p,q} \subset H_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$, every elem in $H^{p,q}$ takes the form $x + yi$, $x, y \in H_{\mathbb{Q}}$ $\rightsquigarrow$ $x - yi \in \overline{H}^{p,q}$)

X cpt Kähler

e.g. $H^k(X, \mathbb{Q})$ has a rational pure Hodge structure of weight k with $H^{p,q} = H^q(X, \Omega_X^p)$

($\overline{H^{p,q}} = H^{q,p}$ by the Dolbeault coh $H^{p,q}_{\bar\partial}(X)$ description)

Equivalent definition using the Hodge filtration:

A <u>rational pure Hodge structure</u> of weight k is a f.d. v.s. $H_{\mathbb{Q}}$ equipped with a decreasing filtration on $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$

$$\cdots \supseteq F^p H_{\mathbb{C}} \supseteq F^{p+1} H_{\mathbb{C}} \supseteq \cdots$$

s.t. $$H_{\mathbb{C}} = F^p H_{\mathbb{C}} \oplus \overline{F^{k-p+1} H_{\mathbb{C}}}, \quad \forall p$$

<u>Exercise</u>. Show two def^{n}s are equivalent:

Given $\{H^{p,q}\} \rightsquigarrow F^p H_{\mathbb{C}} = \bigoplus_{r \geq p} H^{r, n-r}$

Conversely. $H^{p,q} = F^p H_{\mathbb{C}} \cap \overline{F^q H_{\mathbb{C}}}$

e.g. X compact Kähler $\qquad p+q=k$

$\Rightarrow F^p H_{\mathbb{C}} = H^{k-p}(X, \Omega_X^p) \oplus H^{k-p-1}(\Omega_X^{p+1}) \oplus \cdots$

Define $h^{p,q} = \dim_{\mathbb{C}} H^{p,q}_{\bar{\partial}}$ <u>Hodge numbers</u>

Can arrange these into a <u>Hodge diamond</u>:

$$\begin{array}{ccccccc}
 & & & h^{0,0} & & & \\
 & & & \vdots & & & \\
 & & h^{0,n} & \cdots & h^{n,0} & & \\
 & & & \ddots & & & \\
 & & h^{n,n} & & & & \\
\end{array}$$

Fact: The Hodge diamond is symmetric

In part, odd degree coh. must be even-dimensional

e.g. Riemann surface of genus g

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

e.g. complex tori $X = \mathbb{C}^n/\Lambda$

$$h^{p,q} = \binom{n}{p}\binom{n}{q}$$

e.g. $n=3$

$$\begin{array}{ccccccc}
 & & & 1 & & & \\
 & & 3 & & 3 & & \\
 & 3 & & 9 & & 3 & \\
1 & & 10 & & 10 & & 1 \\
 & 3 & & 9 & & 3 & \\
 & & 3 & & 3 & & \\
 & & & 1 & & &
\end{array}$$

Ex / $X = \mathbb{C}^*$

$X \simeq S^1 \Rightarrow H^1(X, \mathbb{Q}) \simeq \mathbb{Q}$

cannot be a pure weight 1 HS

Fact: In Deligne's theory, this carries pure HS of weight 2

Ex / $X = (Y^2Z - X^2(X+Z) = 0) \subset \mathbb{P}^2$

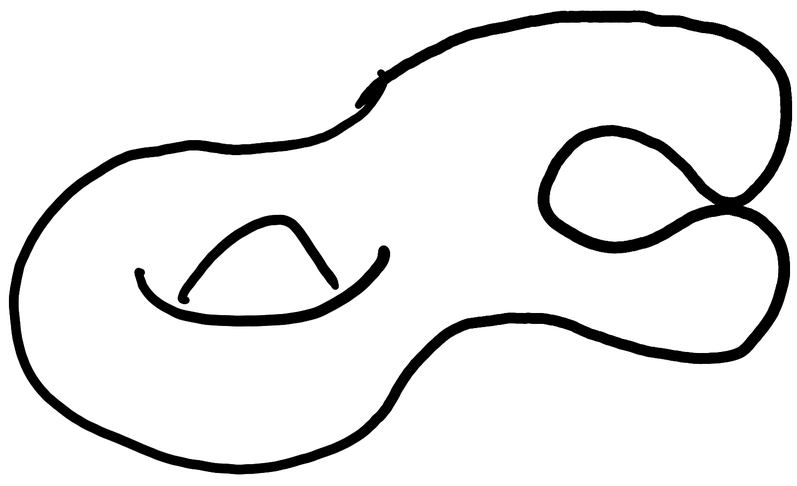
nodal cubic

$\Rightarrow H^1(X, \mathbb{Q}) \simeq \mathbb{Q}$

cannot be a pure weight 1 HS

Fact: In Deligne's theory, this carries pure HS of weight 0

Ex / $X =$ Riemann surface of genus g with p nodes:



$g=1, \quad p=1$

$H^1(X, \mathbb{Q})$

$= \mathbb{Q}^p \oplus \mathbb{Q}^{2g}$

$\mathbb{Q}^p$: pure HS of wt 0

$\mathbb{Q}^{2g}$: pure HS of wt 1

Def^{n}. A <u>rational mixed Hodge structure</u> consists of

① A f.d. $\mathbb{Q}$-v.s. $H_{\mathbb{Q}}$

② An increasing filtration $W_\cdot$ on $H_{\mathbb{Q}}$ — weight filtration

③ A decreasing filtration $F^\cdot$ on $H_{\mathbb{C}}$ — Hodge filtration

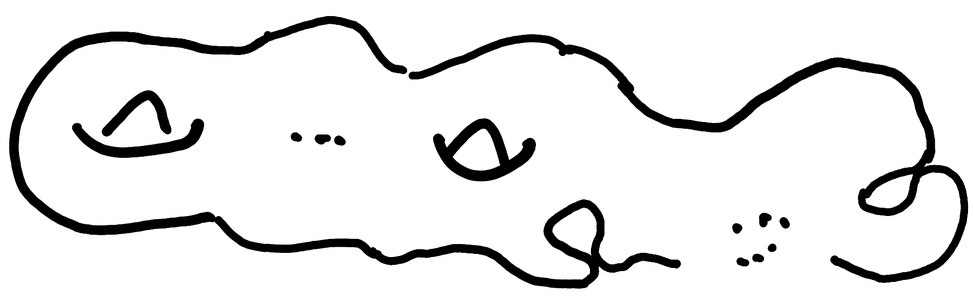
s.t. For each m, the filtration induced by

$F^\cdot$ on $Gr^W_m H_{\mathbb{Q}} = W_m H_{\mathbb{Q}} / W_{m-1} H_{\mathbb{Q}}$

defines a pure HS of weight m

e.g. pure HS of wt k is a MHS with

$$W_{k-1} H = 0 \quad , \quad W_k H = H$$

e.g. $X =$ 

genus g — p nodes

$\Rightarrow H^1(X, \mathbb{Q}) = \mathbb{Q}^{2g} \oplus \mathbb{Q}^p$ carry MHs

with $W_0 H^1 = \mathbb{Q}^p$, $W_1 H^1 = H^1(X, \mathbb{Q})$

$Gr^W_0 H^1 = \mathbb{Q}^p$ — wt 0 pure HS

$Gr^W_1 H^1 = \mathbb{Q}^{2g}$ — wt 1 pure HS

Thm (Deligne)

Let X be a complex algebraic variety
(possibly singular or noncompact)

For every $k \geq 0$, $H^k(X, \mathbb{Q})$ carries a canonical, functorial mixed Hodge structure

References: Deligne's Théorie de Hodge II, III

II: not compact
III: not smooth

We will focus on the compact & singular case
(as the idea is relevant to higher sing. & applications to K-theory)

Nov 6 Deligne's MHS

Recall. A **rational mixed Hodge structure** consists of

① A f.d. $\mathbb{Q}$-v.s. $H_{\mathbb{Q}}$

② An increasing filtration $W_\cdot$ on $H_{\mathbb{Q}}$ (weight filtration)

③ A decreasing filtration $F^\cdot$ on $H_{\mathbb{C}}$ (Hodge filtration)

s.t. For each m, the filtration induced by

$F^\cdot$ on $\mathrm{Gr}^W_m H_{\mathbb{Q}} = W_m H_{\mathbb{Q}} / W_{m-1} H_{\mathbb{Q}}$

defines a pure HS of weight m

e.g. $H^1(\mathbb{C}^*, \mathbb{Q})$ pure HS of wt 2

$H^1(\text{nodal cubic in } \mathbb{P}^2, \mathbb{Q})$ pure HS of wt 0

$H^1(\text{genus } g \text{ curve with } p \text{ nodes}, \mathbb{Q}) \cong \mathbb{Q}^{2g} \oplus \mathbb{Q}^p$

($\mathbb{Q}^{2g}$: pure HS of wt 1; $\mathbb{Q}^p$: pure HS of wt 0)

In general, $H^k(X, \mathbb{Q})$ has
- wt $\geq k$ if X is smooth
- wt $\leq k$ if X is compact

(k to $2k$)

Construction of MHS for X smooth, quasi-proj.

Choose compactification $\bar{X}$ with $D = \bar{X} \setminus X$ SNC

(Ok by Hironaka's thm)

$H^k(X, \mathbb{C})$ is computed by the Logarithmic de Rham

Complex:

$$\mathcal{O}_{\bar{X}} \xrightarrow{d} \Omega^1_{\bar{X}}(\log D) \xrightarrow{d} \Omega^2_{\bar{X}}(\log D) \to \cdots \to \Omega^n_{\bar{X}}(\log D)$$

Locally $D = (z_1 z_2 \cdots z_r = 0)$

$\Omega^1_{\bar{X}}(\log D)$ generated by $\frac{dz_1}{z_1}, \frac{dz_2}{z_2} \cdots \frac{dz_r}{z_r}$

$\Omega^n_{\bar{X}}(\log D) \cong \omega_{\bar{X}}(D)$ generated by $\frac{dz_1 \wedge dz_2 \wedge \cdots \wedge dz_n}{z_1 \cdots z_r}$

Fact: $Rj_* \mathbb{C}_X \cong \Omega^\bullet_{\bar{X}}(\log D)$, where $j: X \to \bar{X}$ is the natural inclusion

Thus $H^k(X, \mathbb{C}) \cong \mathbb{H}^k(\bar{X}, Rj_* \mathbb{C}_X)$

$= \mathbb{H}^k(\bar{X}, \Omega^\bullet_{\bar{X}}(\log D))$

$\uparrow$

filtration on $\Omega^\bullet_{\bar{X}}(\log D)$ induces filtration on $H^k(X, \mathbb{C})$

Define Hodge filtration on $\Omega_{\bar{X}}^{\cdot}(\log D)$ by

$$F^p \Omega_{\bar{X}}^{\cdot}(\log D) = [0 \to \cdots \to 0 \to \Omega_{\bar{X}}^{p}(\log D) \to \Omega_{\bar{X}}^{p+1}(\log D) \to \cdots \to \omega_X(D)]$$

This induces filtration on coh.

$$F^p H^k(X, \mathbb{C}) = \mathrm{Im}\left(\mathbb{H}^k(F^p \Omega_{\bar{X}}^{\cdot}(\log D)) \to H^k(X, \mathbb{C}) \right)$$

The spectral seq

$$E_1^{p,q} = H^q(\bar{X}, \Omega_{\bar{X}}^{p}(\log D)) \Rightarrow H^{p+q}(X, \mathbb{C})$$

degenerates at E_1, so

$$Gr_F^p H^k(X, \mathbb{C}) \cong E_\infty^{p,k-p} \cong E_1^{p,k-p}$$

$\cong$: general theory of spectral seq

$\cong$: deg. at E_1

$$= H^{k-p}(\bar{X}, \Omega_{\bar{X}}^{p}(\log D))$$

e.g. $X = \mathbb{C}^*$, $\bar{X} = \mathbb{P}^1$, $D = \{0, \infty\}$

Logarithmic de Rham complex

$$\Omega^\cdot_{\bar{X}}(\log D) = \left[\mathcal{O}_{\mathbb{P}^1} \xrightarrow{d} \Omega^1_{\mathbb{P}^1}(\log D) \right]$$

$\mathcal{O}_{\mathbb{P}^1}$

($\frac{ds}{s} = -\frac{dt}{t}$ is globally defined)

$$F^0\Omega^\cdot_{\bar{X}}(\log D) = \left[\mathcal{O}_{\mathbb{P}^1} \xrightarrow{d} \mathcal{O}_{\mathbb{P}^1} \right]$$
$$F^1\Omega^\cdot_{\bar{X}}(\log D) = \left[0 \to \mathcal{O}_{\mathbb{P}^1} \right]$$

E_1 page:

$$E_1^{0,1} = H^1(\mathcal{O}_{\mathbb{P}^1}) \to E_1^{1,1} = H^1(\mathcal{O}_{\mathbb{P}^1})$$
$$\| \qquad\qquad\qquad\qquad 0 \| \qquad\qquad\qquad\qquad$$

$$E_1^{0,0} = H^0(\mathcal{O}_{\mathbb{P}^1}) \xrightarrow{0} E_1^{1,0} = H^0(\Omega^1_{\mathbb{P}^1}(\log D))$$
$$\| \qquad\qquad\qquad\qquad \| \qquad\qquad\qquad\qquad$$
$$\mathbb{C} \qquad\qquad\qquad\qquad H^0(\mathcal{O}_{\mathbb{P}^1}) = \mathbb{C}$$

$$\Rightarrow \quad H^0(X, \mathbb{C}) \simeq \mathbb{C} \qquad gr_0^F H^0 = \mathbb{C}$$
$$H^1(X, \mathbb{C}) \simeq \mathbb{C} \qquad gr_1^F H^1 = \mathbb{C}$$

should be (1,1) part in a wt 2 Hs

Define weight filtration

$$W_m \Omega^\cdot_{\bar{X}}(\log D) = \{ \text{log forms with} \le m \text{ denominators} \}$$

e.g. $\frac{dz_1}{z_1} \in W_1 \setminus W_2$, $\frac{dz_1}{z_1} \wedge \frac{dz_2}{z_2} \wedge dz_3 \in W_2 \setminus W_3$

The residue map $\quad gr^W_m \Omega^\cdot_{\bar{X}}(\log D) \xrightarrow{\sim} \Omega^\cdot_{D^{[m]}}[-m]$

$$\frac{dz_{i_1}}{z_{i_1}} \wedge \cdots \wedge \frac{dz_{i_m}}{z_{i_m}} \wedge \omega \mapsto \omega\big|_{z_{i_1} = \cdots = z_{i_m} = 0}$$

where $D = \cup E_i$

is a quasi-isom

$$D^{[m]} = \coprod E_{i_1} \cap \cdots \cap E_{i_m}$$

This induces a filtration on $H^k(X, \mathbb{C}) \cong \mathbb{H}^k(\bar{X}, \Omega^\cdot_{\bar{X}}(\log D))$.

Computed by

$$E_1^{-p,q} = H^{-p+q}\left(gr^W_p \Omega^\cdot_{\bar{X}}(\log D)\right) \Rightarrow H^{-p+q}(X, \mathbb{C})$$

$$\Downarrow$$

$$H^{-p+q}\left(\Omega^\cdot_{D^{[p]}}[-p]\right)$$

$$\Downarrow$$

$$H^{-2p+q}\left(D^{[p]}, \mathbb{C}\right)$$

In fact,

shift the Hodge type (a, b) to $(a+p, b+p)$

$$ {}_wE_1^{-p,q} = H^{-2p+q}\left(D^{[p]}, \mathbb{Q}\right)^{(-p)} \Rightarrow H^{-p+q}(X, \mathbb{Q}) \quad (*) $$

defines the weight filtration on $H^{-p+q}(X, \mathbb{Q})$

This looks like

$$ \cdots \quad \underset{\text{pure wt } 4}{H^0(D^{[2]})(-2)} \xrightarrow{d_1} \underset{\text{pure wt } 4}{H^2(D^{[1]})(-1)} \xrightarrow{d_1} H^4(\bar{X}) $$

$$ \underset{\text{wt } 3}{H^1(D^{[1]})(-1)} \xrightarrow{d_2 = 0} H^3(\bar{X})^{\text{wt } 3} $$

$$ {}_wE_1^{-1,2} = H^0(D^{[1]}) \to H^2(\bar{X}) $$

$$ H^1(\bar{X}) $$

$$ {}_wE_1^{0,0} = H^0(\bar{X}) $$

Exercise. d_1 identifies with the Gysin map (dual to the natural restriction $H^*(D^{[p]}) \to H^*(D^{[p+1]})$) and respects the HS

This degenerates at E_2

(Idea: E_1 differentials are morphisms btw pure HS up to Tate twist of same weight; E_2 differentials are morphisms btw pure HS of different wts, hence zero)

Nov 9

Last time: MHS on X smooth, quasi-proj.

Take $\bar{X}$ = compactification of X

$D = \bar{X} \setminus X$ SNC

Then $F^p \Omega^\cdot_{\bar{X}}(\log D) = \Omega^{\geq p}_{\bar{X}}(\log D)$

$$W_m \Omega^\cdot_{\bar{X}}(\log D) = \sum_{i=0}^{m} \Omega^{p-i}_{\bar{X}} \wedge \Omega^i_{\bar{X}}(\log D)$$

$$= \{ \text{log forms with} \leq m \text{ denominators} \}$$

induce Hodge and weight filtration on

$$H^k(X, \mathbb{C}) = \mathbb{H}^k\left(\Omega^\cdot_{\bar{X}}(\log D)\right)$$

The weight filtration is actually defined over $\mathbb{Q}$, on

$$K_{\mathbb{Q}} = Rj_* \mathbb{Q}_X$$

by $W_m K_{\mathbb{Q}} = T_{\leq m} Rj_* \mathbb{Q}_X$

where $T_{\leq m}$ is the truncation

$$T_{\leq m}\left(\cdots \to C^{m-2} \to C^{m-1} \to C^m \xrightarrow{d} C^{m+1} \to \cdots \right)$$

$$:= \left(\cdots \to C^{m-2} \to C^{m-1} \to \ker d \to 0 \to \cdots \right)$$

So $Gr_m^W K_{\mathbb{Q}} \cong R^m j_* \mathbb{Q}_X[-m]$

We have a filtered quasi-isomorphism

$$(Rj_* \mathbb{Q}_X, W) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} (\Omega_X^\cdot(\log D), W)$$

i.e. ① We have quasi-isom.

$$Rj_* \mathbb{Q}_X \otimes_{\mathbb{Q}} \mathbb{C} \cong Rj_* \mathbb{C}_X \xrightarrow{\sim} \Omega_X^\cdot(\log D)$$

② For each m,

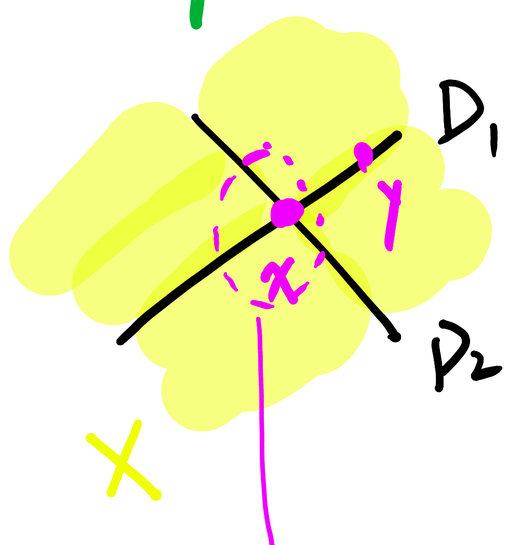
$$Gr_m^W K_{\mathbb{Q}} \otimes \mathbb{C} \xrightarrow{\sim} Gr_m^W \Omega_X^\cdot(\log D)$$

$\S 11$ $\S 11$

$R^m j_* \mathbb{Q}_X[-m] \otimes \mathbb{C}$ $\qquad$ $\Omega_{D^{[m]}}^\cdot[-m]$

identifies with $\mathbb{Q}_{D^{[m]}}$

Claim: $R^m j_* \mathbb{Q}_X \cong \mathbb{Q}_{D^{[m]}}$

e.g. 

$D = D_1 \cup D_2$

- $R^0 j_* \mathbb{Q}_X = \mathbb{Q}_{\bar{X}}$
- $(R^1 j_* \mathbb{Q}_X)_x = H^1(S^1 \times S^1, \mathbb{Q})$
  $\cong \mathbb{Q}^2 \cong \mathbb{Q}_{D_1} \oplus \mathbb{Q}_{D_2}$
  $(R^1 j_* \mathbb{Q}_X)_y = H^1(S^1, \mathbb{Q}) \cong \mathbb{Q} \cdot \mathbb{Q}_{D_2}$
- $R^2 j_* \mathbb{Q}_X \cong \mathbb{Q}_{D_1 \cap D_2}$

Upshot: ① $W_m Rj_* \mathbb{Q}_X = \tau_{\leq m} Rj_* \mathbb{Q}_X$

induces weight filtration on $H^k(X, \mathbb{Q})$

computed by

$${}_wE_1^{-p,q} = H^{-2p+q}(D^{[p]}, \mathbb{Q}) \Rightarrow H^{-p+q}(X, \mathbb{Q})$$

② On $Rj_* \mathbb{Q}_X \otimes_{\mathbb{Q}} \mathbb{C} \simeq \Omega_{\bar{X}}^{\cdot}(\log D)$,

$F^p \Omega_{\bar{X}}^{\cdot}(\log D) = \Omega_{\bar{X}}^{\geq p}(\log D)$ induces

Hodge filtration on $H^*(X, \mathbb{C})$

computed by

$$E_1^{p,q} = H^q(\bar{X}, \Omega_{\bar{X}}^p(\log D)) \Rightarrow H^{p+q}(X, \mathbb{C})$$

③ $F^{\cdot}$ induced on each

$$Gr_m^W Rj_* \mathbb{Q}_X \cong \mathbb{Q}_{D^{[m]}}[-m](-m)$$

is pure of wt m

Ex/ $X = \mathbb{C}^*$ $\quad \overline{X} = D^{[0]} = \mathbb{P}^1$

$D^{[1]} = D = \{0, \infty\}$

E_1 page of wt spectral seq:

$$Q^2 = H^0(\{0,\infty\}) \xrightarrow{d} H^2(\mathbb{P}^1) = Q$$

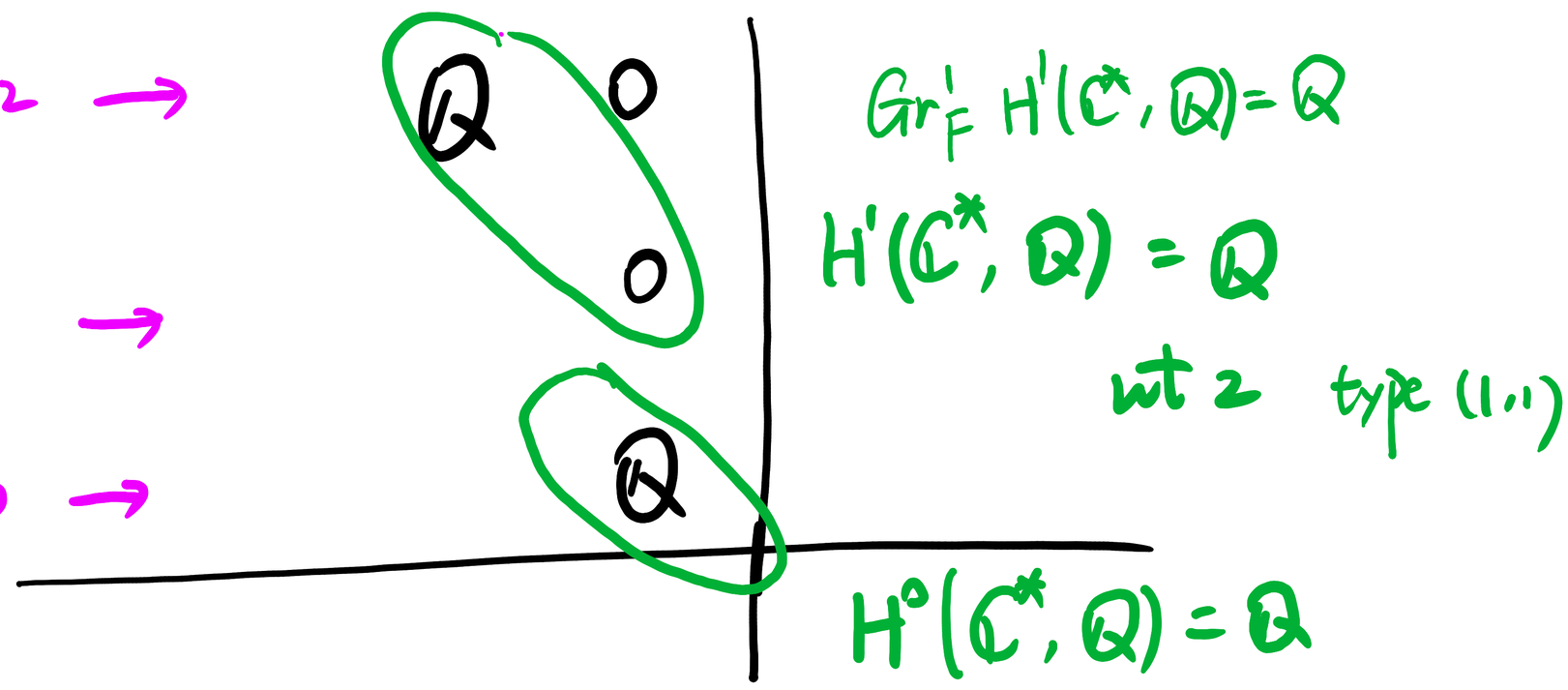
$$H^1(\mathbb{P}^1)$$

$$H^0(\mathbb{P}^1) = Q$$

E_2 page: wt 2 →

wt 1 →

wt 0 →



Exercise $X = C - \{r \text{ points}\}$ where C = genus g curve

$$\Rightarrow H^1(X, Q) = \underbrace{Q^{2g}}_{\text{wt 1}} \oplus \underbrace{Q^{r-1}}_{\text{wt 2}}$$

$h^{0,1} = h^{1,0} = g$

$h^{2,0}$	$h^{1,1}$	$h^{0,2}$
$\|\|$	$\|\|$	$\|\|$
0	$r-1$	0

Construction of MHS for X proj., possibly singular

Take a hyperresolution $X_\bullet \to X$

i.e. a semisimplicial set

$$X_\bullet = \left(\cdots \to X_2 \;\substack{\rightarrow\\\rightarrow\\\rightarrow}\; X_1 \rightrightarrows X_0 \right)$$

s.t.
- All X_i are smooth
- All maps are proper
- $\mathbb{Z}_X \xrightarrow{\sim} R\varepsilon_* \mathbb{Z}_{X_\bullet}$ is qisom

e.g. X = nodal cubic $\subset \mathbb{P}^2$

$$\begin{array}{ccc} \cdots & \to & \mathbb{P}^1 = \\ \downarrow & & \downarrow \\ \bullet & \to & X = \nabla \end{array}$$

$$X_\bullet = \left(X_1 \rightrightarrows X_0 \right)$$
$$\begin{array}{ccc} \| & & \| \\ \vdots & & \mathbb{P}^1 \amalg \bullet \end{array}$$

$= \mathrm{Cone}\left(R\varepsilon_* \mathbb{Z}_{X_1} \to R\varepsilon_* \mathbb{Z}_{X_2} \right)[1]$

The condition $\mathbb{Z}_X \xrightarrow{\sim} R\varepsilon_* \mathbb{Z}_{X_\bullet}$ holds bc we have

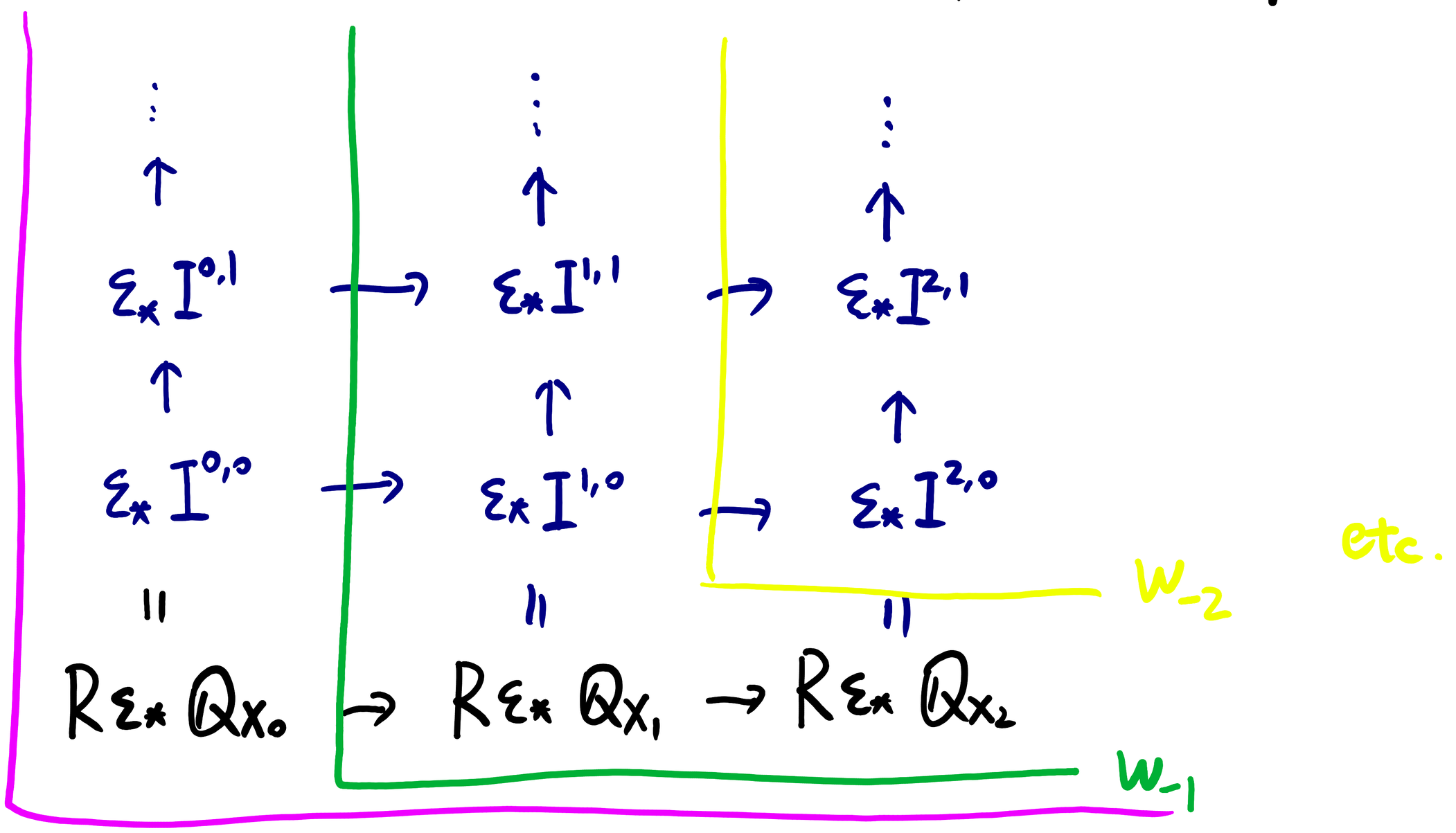
Mayer-Vietoris $\cdots \to H^i(X) \to H^i(\mathbb{P}^1) \oplus H^i(\bullet) \to H^i(\bullet\bullet) \to \cdots$

Since $\mathbb{Q}_X \to R\varepsilon_* \mathbb{Q}_{X.}$,

$$H^k(X, \mathbb{Q}) = \mathbb{H}^k(X, \underbrace{R\varepsilon_* \mathbb{Q}_{X.}}_{\text{put a weight filtration here}})$$

and a Hodge filtration on $R\varepsilon_* \mathbb{C}_X$.

$R\varepsilon_* \mathbb{Q}_{X.}$ is by def^n represented by the complex



$$W_{-k}(R\varepsilon_* \mathbb{Q}_{X.}) = R\varepsilon_* \mathbb{Q}_{X_{\geq k}}$$

$$gr^W_{-k}(R\varepsilon_* \mathbb{Q}_{X.}) = R\varepsilon_* \mathbb{Q}_{X_k}[-k]$$

weight SS: $E_1^{p,q} = H^{p+q}(gr^W_{-p} R\varepsilon_* \mathbb{Q}_{X.}) \Rightarrow H^{p+q}(X, \mathbb{Q})$

$$\overset{q\text{-th}}{H^q(X_p, \mathbb{Q})} \quad \text{deg. at } E_1$$

Hodge filtration

$$F^p(R\varepsilon_* \mathbb{C}_{X.}) = R\varepsilon_* \Omega_{X.}^{\geq p}$$

$$Gr_F^p(\quad) = R\varepsilon_* \Omega_{X.}^{p}[-p]$$

Hodge-to-de Rham spectral seq:

$$E_1^{p,q} = H^{p+q}(Gr_F^p(\quad)) \Rightarrow H^{p+q}(X, \mathbb{C})$$

$$\| $$

$$H^q(X, R\varepsilon_* \Omega_{X.}^p)$$

The p-th Du Bois complex of X denoted $\underline{\Omega}_X^p$

Degeneration at E_1:

$$d: H^q(X, \underline{\Omega}_X^p) \xrightarrow{d} H^q(X, \Omega_X^{p+1})$$

is zero because all $H^q(X_k, \Omega_{X_k}^p) \xrightarrow{d} H^q(X_k, \Omega_{X_k}^{p+1})$ are zero

Ex/ $X = \alpha \subset \mathbb{P}^2$

$X_0 = \mathbb{P}^1 \amalg \cdot$

$X_1 = \cdot\cdot$

pure, wt = 0

$H^1(X, \mathbb{Q}) = \mathbb{Q}$

$W_0 H^1(X, \mathbb{Q}) = \mathbb{Q}$

weight SS, E_1 page:

$\mathbb{Q} = H^2(X_0)$

$0 = H^1(X_0) \longrightarrow H^1(X_1) = 0$

$\mathbb{Q}^2 = H^0(X_0) \longrightarrow H^0(X_1) = \mathbb{Q}^2$

$\mathbb{Q}$ H^2 wt 2 $h^{1,1} = 1$

E_2 page: 0 0

$\mathbb{Q}$ $\mathbb{Q}$

H^0 H^1

wt 0 wt 0

$h^{0,0} = 1$ $h^{0,0} = 1$

$gr_0^F H^1(X, \mathbb{Q}) = H^1(\Omega_X^0) = \mathbb{Q}$

$gr_1^F H^1(X, \mathbb{Q}) = H^0(\Omega_X^1) = 0$

Ex/ $X = \swarrow \simeq S^1$ "rational homology manifold"

$$\begin{array}{c} \mathbb{P}^1 \\ \downarrow \\ X \end{array} \qquad \begin{array}{ccc} X_0 & \to & X \\ \| & & \\ \mathbb{P}^1 & & \end{array} \quad \text{is a hyperres}$$

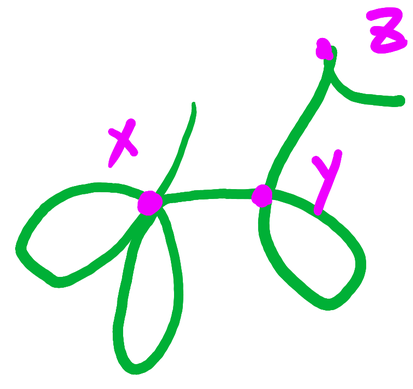
$\Rightarrow H^*(\swarrow, \mathbb{Q}) \cong H^*(\mathbb{P}^1, \mathbb{Q})$ has pure HS

Ex/ C = complex projective irreducible curve

r_x = # branches through x

$\{x_1, x_2, x_3, y_1, y_2, z'\}$

$$\begin{array}{ccc} \| & & \\ E & \longrightarrow & \tilde{C} = \\ \downarrow & & \downarrow \\ C_{sing} & \to & C = \\ \| & & \\ \{x, y, z\} & & \end{array}$$



$r_x = 3$
$r_y = 2$
$r_z = 1$

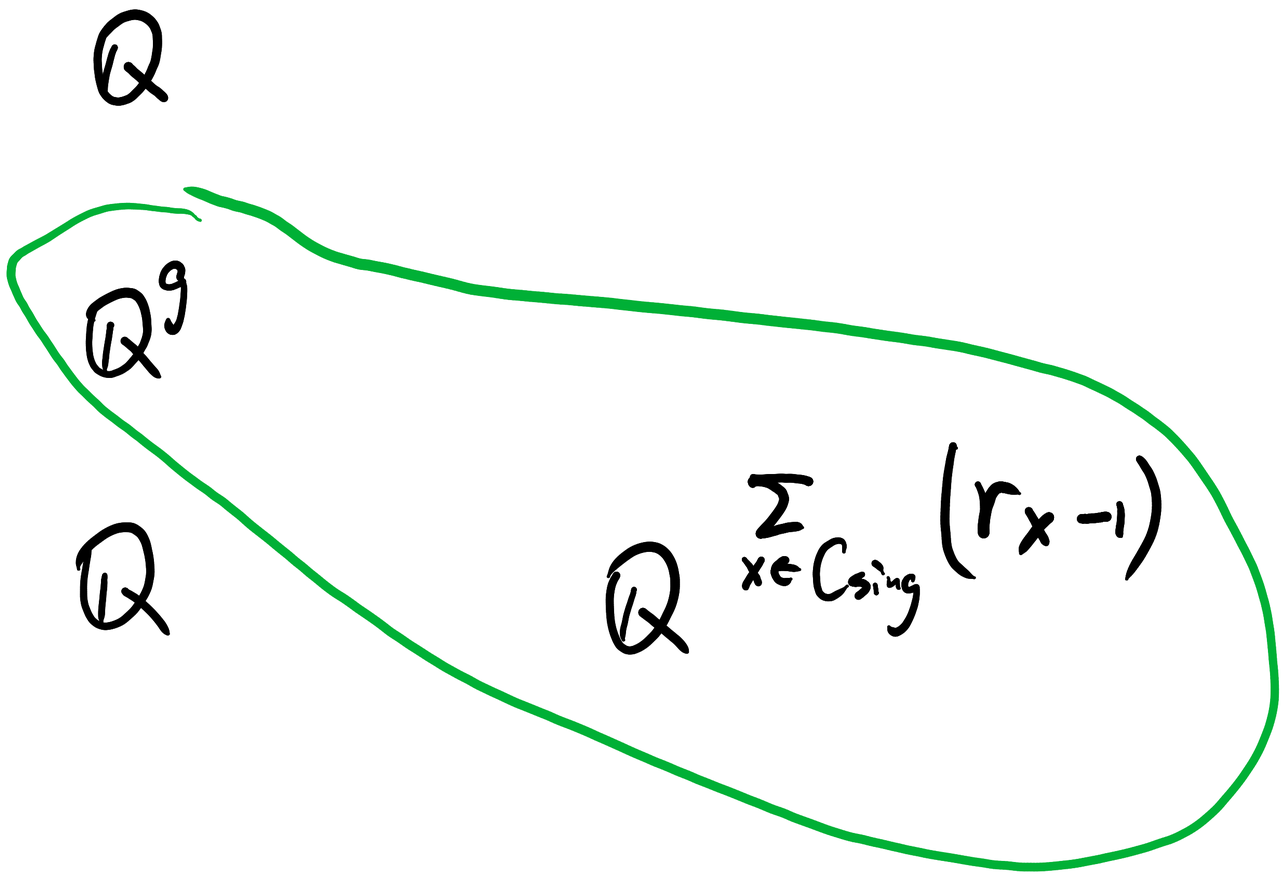
hyperresolution: $\begin{array}{ccc} X_1 & \rightrightarrows & X_0 \\ \| & & \| \\ E & & \tilde{C} \amalg C_{sing} \end{array}$

weight SS, E_1 page $\mathbb{Q} = H^2(X_0) \to H^2(X_1) = 0$

$\mathbb{Q}^g = H^1(X_0) \to H^1(X_1) = 0$

$\mathbb{Q} \oplus \mathbb{Q}^{\#C_{sing}} = H^0(X_0) \twoheadrightarrow H^0(X_1) = \mathbb{Q}^{\sum_{x \in C_{sing}} r_x}$

E_2 page :



$$H^1(C,\mathbb{Q}) = \underbrace{\mathbb{Q}^g}_{\text{wt 1}} \oplus \underbrace{\mathbb{Q}^{\sum_{x\in C_{sing}}(r_x-1)}}_{\text{wt 0}}$$

$$\Rightarrow W_0 H^1(C,\mathbb{Q}) = \mathbb{Q}^{\sum_{x\in C_{sing}}(r_x-1)}$$

$$= \uparrow K_{-1}(C)$$

recall